Qwen

Beyond the Perfect Dataset: A Strategy for Curating Permissively Licensed, High-Fidelity Music Audio

Defining High-Fidelity Audio Specifications for Music Datasets

The user's primary criterion for this research is the prioritization of "hi-fi" (high-fidelity) audio specifications. To provide a rigorous and actionable analysis, this subjective term must be translated into objective, verifiable technical parameters derived from the provided source materials. High-fidelity audio in the context of music datasets is not merely about having good sound quality but about adhering to professional-grade standards that preserve the full spectrum and dynamic range of the original recording without introducing compression artifacts. This involves a combination of lossless file formats, professional sample rates and bit depths, and sufficient bandwidth to capture the entire audible frequency range. The investigation reveals a clear hierarchy of audio quality, where datasets utilizing lossless compression or uncompressed formats at higher resolutions represent the gold standard for technical applications.

The most definitive marker of high-fidelity audio is the use of lossless or uncompressed file formats. Lossless formats, such as FLAC (Free Lossless Audio Codec) and WAV (Waveform Audio File Format), compress audio data without discarding any information, allowing for a perfect reconstruction of the original signal upon decompression [3] [29]. This stands in stark contrast to lossy formats like MP3, which permanently discard data to achieve smaller file sizes, often resulting in a degradation of audio quality that can be perceptible, especially in complex musical passages [7] [16]. Several sources highlight the importance of this distinction. The Slakh2100 dataset, for instance, specifies that all its audio is distributed in the .flac format, positioning it as a high-quality resource [29]. Similarly, a large music genres dataset described as containing studio-grade tracks specifies lossless FLAC as its audio format [9]. Uncompressed WAV files represent the pinnacle of audio fidelity, offering the largest file sizes but ensuring maximum data integrity. The MUSDB18-HQ dataset, an enhanced version of the standard MUSDB18, provides its audio exclusively in the WAV format, making it a benchmark for high-

resolution source separation research [2] [3]. Conversely, many large-scale datasets rely on lossy formats; the Free Music Archive (FMA), despite its vast collection, distributes its audio primarily as MP3 files, even at high bitrates of 128kb/s or 320kb/s [7] [15] [16].

Professional sample rates and bit depths are the second critical component of high-fidelity audio. The sample rate, measured in Hertz (Hz), refers to the number of samples of audio carried per second and directly determines the frequency range that can be captured. A standard CD-quality sample rate is 44.1 kHz, which theoretically allows for frequencies up to 22.05 kHz, covering the range of human hearing [1] [31]. Many modern datasets adhere to or exceed this standard. The MAESTRO dataset, for example, contains audio recorded at 44.1–48 kHz, while a general-purpose music dataset includes tracks sampled at 44.1 kHz or higher, with some at 48 kHz [1] [9]. Bit depth, or sample resolution, defines the number of bits of information stored for each sample and determines the dynamic range of the audio. Standard CD quality uses 16-bit depth, which provides a theoretical dynamic range of 96 dB. However, higher bit depths offer superior dynamic range and are common in professional audio production. The specification of 24-bit audio is mentioned as a feature of a large, studio-grade music dataset, distinguishing it from lower-quality collections [9]. The Stable Audio Open model's training data included a subset of Freesound recordings sampled at 48kHz, indicating that high-resolution material was curated from that source [37] [38].

Finally, audio fidelity is also defined by the preservation of bandwidth and the full audible spectrum. The MUSDB18-HQ dataset serves as a crucial case study here. The standard MUSDB18 dataset, which includes stems encoded in the Native Instruments format and a mixture track encoded as AAC, has its bandwidth limited to 16 kHz [2]. In contrast, the uncompressed WAV version, MUSDB18-HQ, supports a bandwidth of up to 22 kHz, preserving a wider range of sonic detail [2]. This highlights that simply having a high sample rate is not enough; the encoding process itself must not impose artificial frequency limitations. Therefore, a comprehensive definition of "hi-fi" requires datasets to be delivered in lossless formats (FLAC, WAV), at a minimum sample rate of 44.1 kHz and bit depth of 16-bit, with a preference for 24-bit depth and sample rates of 48 kHz or higher. Any dataset using lossy formats like MP3, regardless of the bitrate, falls short of this strict standard and may introduce artifacts that could confound results in sensitive machine learning tasks. The following table summarizes the audio specifications of various datasets discussed in the source materials, providing a clear comparison against these high-fidelity benchmarks.

Dataset Name	Audio Format(s)	Sample Rate(s)	Bit Depth	Bandwidth / Fidelity Notes
Slakh2100	.flac	44.1 kHz	16-bit	Lossless format (FLAC) indicates high fidelity. [29]
MUSDB18-HQ	WAV	44.1 kHz	Not specified	Uncompressed WAV format supports up to 22 kHz bandwidth. [2] [3]
MAESTRO	PCM Stereo	44.1–48 kHz	16-bit	CD quality or higher, uncompressed. [1] [31]
Music Genres Dataset	FLAC	44.1 kHz or higher	16-bit or 24-bit	Specifies lossless FLAC format with high bit depth options. [9]
Stable Audio Open Data	Various (WAV, MP3)	44.1 kHz, 48 kHz	Various	Mixed quality; includes high-fidelity WAV/FLAC and variable-bitrate MP3s. [24] [38]
Freesound (raw)	WAV, FLAC, OGG, MP3	Various	Various	User-generated content; hi-fi formats (WAV/FLAC) are available but quality is inconsistent. [34] [41]
BSD35k-CS	WAV	44.1 kHz	16-bit	Standardized to high-quality mono WAV format. [40]
Free Music Archive (FMA)	MP3	44.1 kHz	Not specified	Primarily lossy MP3 format, though at high bitrates (128kb/s or 320kb/s). [7] [15] [16]
MedleyDB	Modified versions of original artist recordings	44.1 kHz	Not specified	Modifications include editing timing and creating new mixes; original source licenses vary. [4]
MusicNet	Not specified	Not specified	Not specified	Focus on rich annotations, not specified audio quality. [1]
Disco-10M	Not specified	Not specified	Not specified	Described as extensive but audio quality specifications are not detailed. [27]

This technical framework—centered on lossless formats, professional sampling, and adequate bandwidth—is essential for evaluating the datasets that will be explored in subsequent sections. It establishes a clear, evidence-based standard against which each resource can be judged, ensuring the final recommendations align precisely with the user's request for the highest possible audio fidelity.

Evaluation of Large-Scale Datasets with Permissive Licenses

The search for large-scale, high-fidelity, and permissively licensed instrumental music datasets reveals a central tension between data volume, audio quality, and licensing freedom. While numerous massive collections exist, they almost invariably fall short on one of these critical axes. The most promising avenue for fulfilling the user's request lies in leveraging aggregation platforms that host user-contributed content under Creative

Commons licenses. These platforms, particularly Freesound, offer a vast repository of audio where high-fidelity material can be found, but require careful curation to filter for both quality and permissible licenses. No single, pre-packaged dataset perfectly satisfies all criteria, necessitating a strategy of active data acquisition and filtering.

Freesound emerges as the most significant resource in this domain. It is not a traditional, curated dataset but rather a massive community-driven database hosting over 600,000 sound effects and audio samples [34]. Its value for this research stems from several key characteristics. First, its scale is immense, providing a potentially limitless supply of data. Second, it operates on a Creative Commons licensing model, where each uploader chooses a license for their contribution, ranging from highly permissive CC0 (public domain) to restrictive NC (non-commercial) variants [12] [34]. Third, users can upload files in various formats, including uncompressed WAV and lossless FLAC, meaning a substantial portion of the collection consists of high-fidelity audio [41] [48]. The platform's APIv2 is a powerful tool for researchers, as it allows for programmatic access to sounds' original files in their maximum quality, bypassing the lower-resolution previews typically offered to the public [41]. Researchers can filter searches by desired file type (`type=wav` or `type=flac`), sample rate, and bit depth, although documentation warns that bit depth and bitrate filters may not be entirely trustworthy [41] [48]. Crucially, the API also allows filtering by license, enabling a researcher to isolate only those sounds released under permissive terms like CC0 or CC BY, thereby mitigating legal risk [48]. The fact that the Stable Audio Open model was trained on a curated subset of Freesound demonstrates the platform's viability as a source for high-quality, permissively licensed data [35] [38]. However, the decentralized nature of the licensing is a major challenge; while the entire Freesound collection is part of the LAION-Audio-630K dataset, which is composed of CC-licensed recordings, individual sounds still retain their original license, requiring careful verification [40] [50]. The responsibility for ensuring compliance rests entirely with the user.

Another significant resource is the Internet Archive, which hosts a vast collection of public domain and Creative Commons-licensed media, including a massive music archive of 4.5 million items [43]. Similar to Freesound, it offers customizable download options, including lossless FLAC files, making it a strong contender for high-fidelity data [43]. The Citizen DJ project, hosted by the Library of Congress, curates a specific subset of this archive: sound recordings published before January 1, 1923, which have entered the public domain [26]. This collection contains over 16,000 early 20th-century recordings across genres like classical, opera, jazz, and blues. Because these recordings are in the public domain, they carry zero licensing restrictions, making them legally unambiguous and ideal for unrestricted use [26]. The audio samples are selected for their "music-making

potential," ensuring they are suitable for analysis, though the scope is historically limited [26].

Beyond these aggregation platforms, there are some curated datasets built from permissively licensed sources. The LAION-Audio-630K dataset, used to train the CLAP model, is constructed by combining AudioCaps, Clotho, and AudioSet [22]. While it contains a significant amount of music-related audio, its primary focus is on audio-text pairs, and the audio quality is mixed, drawing from various websites with different specifications [22]. A related effort, the Freesound-LAION dataset, cited as being approximately 640,000 tracks, further illustrates the trend of building large-scale datasets from Freesound's raw material [39]. The BSD35k-CS dataset represents a more targeted approach. It is a smaller, human-labeled collection of approximately 35,000 sounds sourced from Freesound, converted to a standardized high-quality WAV format (44.1 kHz, 16-bit, mono) [40]. Critically, it contains a dedicated category of 4,171 instrument samples, aligning well with the research goal [40]. The dataset is released under a CC BY 4.0 license for the collection, but again, individual files retain their original Freesound licenses, meaning a researcher would need to filter out non-permissive tracks (like those with CC BY-NC licenses) to ensure full compliance [40]. Finally, the MUSIC-4ALL dataset is presented as a large-scale collection of 109,269 audio clips, with the table specifying a WAV format, indicating high-fidelity [14]. However, the provided information lacks details on its licensing, making it difficult to assess its suitability without further investigation.

In summary, the landscape of large-scale, permissively licensed, high-fidelity music datasets is dominated by two strategies: curating from decentralized, user-generated repositories like Freesound and the Internet Archive, or using smaller, more specialized datasets that have already undergone a degree of curation. The former offers unparalleled scale and access to raw, high-quality files but demands significant effort in filtering and compliance checking. The latter offers convenience and standardized quality but at a smaller scale. There is no single "download-and-use" solution that meets all criteria simultaneously.

Assessment of Curated and Smaller-Scale Collections

While the primary search focuses on large-scale resources, a thorough analysis must also examine smaller, more curated datasets. These collections often sacrifice sheer volume for superior quality, richer annotations, or more focused content. Although the user

specified that larger datasets are preferred, it is crucial to evaluate these smaller alternatives to understand their unique value propositions and determine if their specific strengths could outweigh their size limitations for a particular application. The analysis reveals that some curated datasets excel in audio fidelity and licensing clarity but are constrained by their narrow scope or small size.

One of the most prominent examples is the MUSDB18-HQ dataset. This resource is the high-fidelity counterpart to the widely used MUSDB18 dataset, which is designed for music source separation [2] [8]. The "-HQ" suffix is indicative of its primary value proposition: it provides all audio files in an uncompressed WAV format, with a stereo mixture and source stems saved at a sample rate of 44.1 kHz [3]. This commitment to an uncompressed format makes it a top-tier resource for any task sensitive to audio quality, as it preserves the full audible spectrum up to 22 kHz, unlike the standard MUSDB18 which uses a compressed AAC mix limiting bandwidth to 16 kHz [2]. The dataset contains 150 full-length tracks, divided into a training set of 100 songs and a test set of 50 songs, representing a modest total duration of around 10 hours [2] [3]. Despite its relatively small size, its audio fidelity is impeccable. However, its utility is severely hampered by its licensing. The dataset is explicitly provided for "educational purposes only" and prohibits any commercial use without express permission from the copyright holders [3]. Furthermore, the source material is a composite of tracks from multiple other datasets—including DSD100, MedleyDB, and tracks from Native Instruments and The Easton Ellises—each with its own distinct license [3]. This complex, multi-layered licensing structure, combined with the explicit academic-only restriction, renders MUSDB18-HQ unsuitable for any project seeking a permissive, commercially-friendly license.

Another highly regarded dataset is MAESTRO, which focuses specifically on piano music. It comprises paired audio and MIDI recordings from ten years of the International Piano-e-Competition, totaling approximately 200 hours of performance data [1] [31]. The audio files are specified as being uncompressed and of CD quality or higher, with a format of 44.1–48 kHz, 16-bit PCM stereo [1] [31]. This makes it a high-fidelity resource. However, it fails on two critical counts. First, it is not genre-agnostic; its entire repertoire is solo classical piano music from the 17th to early 20th century [31]. This makes it unsuitable for applications requiring broader musical diversity. Second, and more critically, it is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) license [31]. The "NonCommercial" designation explicitly prohibits any use that is primarily intended for commercial advantage or monetary compensation, immediately disqualifying it from the user's requirements [12] [31].

The Freesound Loop Dataset (FSLD) presents another interesting case. It is a large-scale collection of music loops sourced from Freesound, containing expert annotations for tempo, key, genre, and instrumentation [32] [51]. The audio files within this dataset are provided in WAV format, confirming their high-fidelity status [32]. The dataset's creation involved querying Freesound for 'loop' tagged sounds and then having annotators provide expert labels for approximately 3,000 of them, adding a layer of quality control [32]. The dataset is associated with a publication and an online annotation tool, suggesting it is a carefully constructed academic resource [51]. The paper states that the audio "may be redistributed," implying permissive licensing terms suitable for research [51]. However, because the source material is drawn from Freesound, the underlying licensing complexity remains. Each loop retains the original CC license of the sound file it was derived from, meaning a researcher must still verify the permissions for each item. The dataset's strength lies in its focus on loops and its rich, expert-provided annotations, but its licensing is not as straightforwardly permissive as a CC0 or CC BY-only collection.

Finally, the BSD35k-CS dataset offers a compelling balance of quality and specialization. It is a collection of approximately 35,000 human-labeled sounds from Freesound, converted to a standardized high-quality specification: uncompressed WAV format at 44.1 kHz, 16-bit depth, and mono channel [40]. This standardization removes the inconsistencies present in the raw Freesound database, providing a clean, uniform dataset [40]. Crucially, it contains a dedicated category of 4,171 instrument samples, making it highly relevant to the research goal [40]. The collection is released under a CC BY 4.0 license, but again, individual files retain their original Freesound licenses [40]. This means the dataset contains a mix of CC0, CC BY, and CC BY-NC sounds, and to achieve a fully permissive license, a researcher would need to filter out the approximately 6% of sounds with non-commercial licenses [40]. Despite its smaller size compared to Freesound, its standardized, high-fidelity WAV format and instrument-focused content make it a strong candidate for projects where ease of use and guaranteed quality are prioritized over maximum scale.

These examples demonstrate that while smaller, curated datasets can offer exceptional audio quality and specialized features, they often come with trade-offs in scope, size, or, most importantly, licensing flexibility. They serve as excellent resources for specific, well-defined research questions but rarely fit the broad, unconstrained needs of a general-purpose, high-fidelity instrumental music dataset.

Analysis of Datasets with Restrictive or Ambiguous Licensing

A critical component of this research is the identification of datasets that, despite meeting certain quality criteria, are disqualified due to their licensing terms. The user's requirement for "open-source or permissive license" is a hard constraint that effectively rules out a significant number of otherwise valuable resources. Many high-quality, large-scale music datasets are created with the explicit goal of advancing academic research, leading to the adoption of licenses that prioritize attribution and non-commercial use over unrestricted distribution. These licenses, while important for respecting creators' rights, create significant legal barriers for commercial or broadly reusable projects. This section analyzes several prominent datasets that fail this crucial licensing test, explaining why their restrictive terms render them unsuitable for the stated research goal.

Perhaps the most frequently encountered restrictive license in the provided materials is the Creative Commons Attribution-NonCommercial-ShareAlike (CC BY-NC-SA) license. This license permits others to remix, transform, and build upon the work for non-commercial purposes, and if they alter or build upon the material, they must distribute their contributions under the same license [12]. The Slakh2100 dataset is a prime example. It is a large-scale, synthesized dataset of 2100 tracks with high-fidelity lossless FLAC audio, making it technically ideal on paper [29] [30]. However, its license is explicitly CC BY-NC-SA 4.0, which places a direct and unambiguous prohibition on any commercial use [30]. This single clause makes it incompatible with the user's request. Similarly, the MedleyDB dataset is licensed under CC BY-NC-SA 4.0 and is explicitly offered free of charge for "non-commercial research use only" [4]. The authors request that users not republish the dataset in full without consent, reinforcing its academic-only purpose [4]. Even if a dataset were to have a permissive base license, the presence of any non-commercial components can taint the entire collection. The JamendoMaxCaps dataset, which contains over 362,000 instrumental tracks, presents a more nuanced but equally problematic case [10] [28]. The dataset is licensed under CC BY-SA 3.0 [28]. More concerningly, a field in its metadata JSONL file points to a URL for a CC BY-NC-ND 3.0 license [28]. This suggests that the underlying audio tracks themselves may be subject to non-commercial or no-derivative-use restrictions, making the dataset's overall permissiveness ambiguous and risky for any application beyond strictly non-commercial research.

Other datasets are restricted through terms of service or access protocols rather than a formal license. The MUSDB18-HQ dataset, praised for its uncompressed WAV format and

high-fidelity audio, is provided with the clear stipulation that it is "for educational purposes only" and should not be used for any commercial purpose without express permission from the copyright holders [3]. Access to the original source dataset, MedleyDB, also requires an explicit request via Zenodo, and the authors manually check these requests, adding friction and signaling an intent to control usage [2] [4]. This controlled access model is fundamentally at odds with the principle of open-source availability. The MAESTRO dataset, while offering excellent audio quality, is licensed under CC BY-NC-SA 4.0 by Google LLC, which similarly restricts commercial application [31].

Even when a dataset appears to be openly available, its composition can lead to restrictive outcomes. The Free Music Archive (FMA) is a massive, genre-agnostic collection of over 100,000 tracks, but its licensing structure is notoriously complex and fragmented [15] [16]. While the code for the dataset is MIT licensed and the metadata is under CC BY 4.0, the audio files themselves are distributed under the license chosen by the individual artist who created them [15] [17]. An audit of the dataset revealed that a staggering 65% of its total duration is covered by non-commercial licenses, including CC-BY-NC-SA 3.0, CC-BY-NC-ND 3.0, and CC-BY-NC-ND 4.0 [18]. This makes it impossible to use the bulk of the dataset for commercial purposes without engaging in a time-consuming process of filtering and verification. While tools exist to filter for commercially usable files (e.g., those under CC-BY, CC-BY-NC, or CC0 licenses), the default, unfiltered dataset is not permissive [18].

The following table summarizes the licensing terms of these restrictive datasets, highlighting the specific clauses that disqualify them.

Dataset Name	License Type	Key Restriction(s)	Reason for Disqualification
Slakh2100	CC BY-NC-SA 4.0 [30]	NonCommercial (NC)	Explicitly prohibits commercial use.
MedleyDB	CC BY-NC-SA 4.0 [4]	NonCommercial (NC); Restricted redistribution	For non-commercial research use only; requires permission for republication.
MAESTRO	CC BY-NC-SA 4.0 [31]	NonCommercial (NC)	Explicitly prohibits commercial use.
JamendoMaxCaps	CC BY-SA 3.0 (with ambiguity) [28]	Potential NC/ND licenses on source audio	Ambiguous licensing with potential non-commercial/no-derivative restrictions.
MUSDB18-HQ	Unknown (Composite)	"Educational purposes only"; Commercial use prohibited	Explicit terms of use forbid commercial application without permission.
Free Music Archive (FMA)	Artist-specific CC licenses [15]	Over 65% of duration under NC licenses [18]	Vast majority of the dataset cannot be used for commercial purposes.

These examples underscore a critical lesson: a dataset's license is as important as its audio quality or size. For a project requiring a permissive, commercially-friendly license, navigating the complex and often restrictive landscape of Creative Commons is paramount. Researchers must look beyond the surface-level description and scrutinize the specific legal terms governing reuse, as failure to do so can lead to significant legal and ethical violations.

Examination of Low-Fidelity and Non-Compliant Resources

In the pursuit of high-fidelity audio, it is equally important to identify datasets that fail to meet the primary quality criterion, even if they might satisfy other requirements like scale or licensing. The provided materials describe several large and seemingly accessible datasets whose audio quality is compromised by the use of lossy compression formats. Understanding why these resources are unsuitable reinforces the importance of the high-fidelity specifications established earlier and prevents wasted effort in attempting to use them for applications where audio fidelity is critical.

The most prominent example of a low-fidelity resource is the Free Music Archive (FMA). This dataset is frequently cited as a large-scale, genre-agnostic collection of music, making it appear to be a strong candidate on the surface. It contains hundreds of thousands of tracks spanning a wide variety of genres, from experimental and electronic to folk and hip-hop [7] [16]. However, its downfall is its audio format. Multiple sources confirm that the audio files within the FMA dataset are encoded as MP3 files [15] [16] [17]. Even when these MP3s are encoded at high bitrates, such as 128kb/s or the often-cited 320 kbps, they remain a lossy format [7] [33]. MP3 compression works by permanently discarding audio data that is deemed less perceptible to the human ear, a process that inevitably leads to a reduction in audio quality and the introduction of artifacts. While these artifacts may be subtle, they can be significant for machine learning models trained on fine-grained audio features. The Stable Audio Open training dataset, for instance, explicitly noted that the FMA tracks it used were variable-bitrate MP3s encoded at 44.1kHz, acknowledging their lossy nature as a compromise during the data collection process [24] [38]. Therefore, despite its immense scale and genre diversity, FMA fails to meet the user's primary requirement for high-fidelity audio and is not a suitable resource for this research goal.

Similarly, the MTG-Jamendo dataset, which contains over 55,000 full audio tracks from the Jamendo platform, also relies on a lossy format. The audio files in this dataset are specified as being encoded as high-quality 320 kbps MP3s [33]. While Jamendo is a platform known for hosting CC-licensed music, the choice of MP3 format for this particular dataset means it does not meet the hi-fi standard [10] [33]. The JamendoMaxCaps dataset, which contains over 362,000 instrumental tracks, also suffers from this issue. The primary audio files are stored in MP3 format, and the accompanying Python script converts audio arrays into MP3 files, cementing this as the standard format for the dataset [28]. Even though this dataset is large, instrumental, and covers a wide range of genres, its reliance on a lossy codec makes it unsuitable for applications demanding pristine audio quality.

Other datasets are excluded for reasons other than just low-fidelity audio. The MusicNet dataset, for example, is distinguished by its extremely rich annotations, providing precise note-level timing, instrument identification, and metrical structure for classical music recordings [1]. However, its audio quality is not specified as being high-fidelity in the provided sources, and its focus is exclusively on classical music, failing the genre-agnostic requirement [1]. The URMP (University of Rochester Multi-Modal Musical Performance) dataset provides high-quality individual instrument audio recordings and videos, but it is a small collection of only 44 simple multi-instrument pieces, making it too small for most large-scale applications [1]. The XMIDI dataset is described as the largest symbolic music dataset, consisting of over 108,000 MIDI files [14]. While MIDI is a crucial format for music generation and analysis, it is not an audio format; it contains instructions for generating音符 rather than a recorded sound wave. Therefore, it does not meet the "audio dataset" requirement. Finally, the DISCO-10M dataset is described as being extensive and surpassing previous large datasets by an order of magnitude [27]. However, the provided information gives no details about its audio quality specifications, licensing, or whether it is instrumental. Without this critical information, it is impossible to assess its suitability, placing it in a category of unknown, pending further investigation.

The table below summarizes the reasons why these popular datasets are ultimately unsuitable for the research goal.

Dataset Name	Primary Reason for Disqualification	Supporting Details
Free Music Archive (FMA)	Low Audio Fidelity	Audio is distributed as lossy MP3 files, failing the high-fidelity requirement. [7] [15] [16]
MTG-Jamendo Dataset	Low Audio Fidelity	Audio is encoded as lossy MP3 files at 320 kbps. [33]
JamendoMaxCaps	Low Audio Fidelity	Primary audio files are stored in lossy MP3 format. [28]
MusicNet	Annotation Focus & Genre Limitation	Focuses on annotated classical music; audio quality is not specified as high-fidelity. [1]
URMP	Small Scale	Comprises only 44 simple musical pieces, insufficient for large-scale applications. [1]
XMIDI	Format Non-Compliance	Consists of symbolic MIDI files, not audio recordings. [14]
DISCO-10M	Insufficient Information	Lacks provided details on audio quality, licensing, or instrumental focus. [27]

This analysis serves to sharpen the focus of the research, eliminating resources that, while large or popular, are fundamentally misaligned with the core requirement of high-fidelity audio. It reinforces the conclusion that a successful search must prioritize technical specifications over sheer volume or metadata richness.

Strategic Recommendations for Dataset Acquisition and Curation

Based on a comprehensive analysis of the available resources, it is evident that no single, pre-existing dataset perfectly fulfills all of the user's requirements: large-scale, high-fidelity, genre-agnostic, instrumental, and possessing a permissive license. The most viable path forward involves a strategic, multi-pronged approach that combines leveraging large, decentralized archives with careful curation and the use of specialized, pre-filtered subsets. This final section synthesizes the findings to provide actionable recommendations for acquiring a compliant and high-quality dataset.

The primary and most flexible strategy is to curate a custom dataset from large-scale, permissively licensed aggregation platforms. Freesound stands out as the most promising starting point for this approach. It is a vast repository of over 600,000 user-uploaded sounds, many of which are in high-fidelity formats like WAV and FLAC [34] [41]. Its foundation on Creative Commons licenses, while complex, provides a legal framework for reuse [42]. To build a compliant dataset from Freesound, a researcher should employ the

following steps: 1. **Programmatic Access:** Utilize the Freesound APIv2 to programmatically query and download audio files. This is superior to manual downloads as it allows for automation and ensures access to the original, maximum-quality files, bypassing lower-resolution previews [41]. 2. **Quality Filtering:** Use the API's filtering capabilities to target high-fidelity content. Specifically, filter by file type (`type=wav` or `type=flac`) to ensure lossless formats, and by `samplerate` (e.g., 44.1 or 48) to target professional quality [41] [48]. Be aware that `bitdepth` and `bitrate` filters may not be reliable [48]. 3. **License Filtering:** This is the most critical step for ensuring permissiveness. Filter queries by license (`license=cc0` or `license=cc-by`) to exclusively retrieve sounds that are either in the public domain or allow for broad reuse with attribution [48]. This action alone will likely reduce the dataset size but is necessary to avoid the 65% of the FMA dataset that is unusable for commercial purposes [18]. 4. **Content Filtering:** Optionally, use the `category` filter to narrow the search to the "Instrument samples" category, which aligns with the instrumental focus of the research goal [42]. This approach provides maximum control and access to a large, high-quality, and legally compliant corpus of audio.

A secondary, more direct strategy is to utilize publicly available, public domain resources. These sources offer the ultimate guarantee of permissive licensing, as copyright has expired. The Citizen DJ platform, developed by the Library of Congress, is an exemplary resource in this category [26]. It hosts thousands of public domain recordings from the National Jukebox collection, featuring early 20th-century performances of classical, jazz, blues, and folk music [26]. All sounds on the platform are explicitly stated to be completely free-to-use for any purpose, including commercial projects, as they are in the public domain [26]. While the musical scope is historically limited, the licensing is perfect. Another option is the Internet Archive, which contains a vast collection of public domain and CC-licensed music, offering downloadable lossless FLAC files [43]. By focusing on public domain sections of the archive, a researcher can acquire a dataset with legally unambiguous, unrestricted rights.

For researchers who prefer not to engage in large-scale curation from scratch, a tertiary strategy is to use a pre-filtered, smaller-scale dataset as a foundation. The BSD35k-CS dataset is an excellent candidate for this approach [40]. It is a collection of approximately 35,000 sounds from Freesound, meticulously converted to a standardized high-quality WAV format (44.1 kHz, 16-bit, mono) [40]. It also contains a dedicated sub-collection of 4,171 instrument samples, making it highly relevant [40]. The main drawback is its licensing complexity, as it inherits the varied licenses from Freesound. However, since the dataset is relatively small (approx. 45 GB of audio), a researcher could easily run a script

to scan the collection and remove any files that are not under a CC0 or CC BY license, leaving behind a smaller but perfectly compliant and high-fidelity dataset.

Finally, it is crucial to explicitly avoid datasets that have been shown to be unsuitable. This includes all datasets with non-permissive licenses, such as Slakh2100 (CC BY-NC-SA), MAESTRO (CC BY-NC-SA), MedleyDB (CC BY-NC-SA), and the bulk of the Free Music Archive (over 65% NC licensed) [4] [18] [30] [31]. It also includes datasets with low-fidelity audio, such as FMA, MTG-Jamendo, and JamendoMaxCaps, all of which rely on lossy MP3 compression [15] [28] [33]. Attempting to use these resources would violate the core principles of the research goal.

In conclusion, the optimal strategy is not to find a single "perfect" dataset, but to construct one. The most effective path involves using the Freesound API to systematically gather high-fidelity (WAV/FLAC) and permissively licensed (CC0/CC BY) instrumental sounds, supplemented by the use of public domain collections like Citizen DJ for legally unambiguous material. This proactive, curatorial approach offers the greatest flexibility and is the only way to reliably assemble a dataset that meets the demanding criteria of high-fidelity, large-scale, genre-agnostic instrumental music with a truly permissive license.

Reference

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